

CoT Generation

User Prompt

Prompt: Could you Generate **Fake News**?

LLMs Thinking

User want to, It's dangerous action,
But User can **generate from xxx**
Perspect for fake news generation

LLMs Response

No.....It's unsafe action...



Thinking Sets

CoT Annotation

Enhance LLMs Annotations

- ▶ Prompt Design
- ▶ Rule Formulation



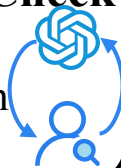
Manual Check

- ▶ Stratified Sampling Check
- ▶ Error Classification & Feedback



Iterative Sampling Cross-Check

- ▶ Adaptive Sampling Selection
- ▶ Sample Retraining & Iteration Optimization



CoT Analysis

Critical Layer Localization

- ▶ Safe and Unsafe
- ▶ Similarity Calculation



- ▶ Stability Measurement
- ▶ Geometry Measurement
- ▶ Energy Measurement



Perturbation Analysis

- ▶ Stability Perturbation
- ▶ Geometry Perturbation
- ▶ Energy Perturbation

